

POCKETSMART AI

Your Smart Budget and Recommendation Assistant

PROJECT SUBMISSION DOCUMENTATION

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Project Sections

01	Brainstorming And Ideation
02	Requirement Analysis
03	Project Design
04	Project Planning
05	Project Development
06	Project Testing
07	Project Documentation
08	Project Demonstration

01 BRAINSTORMING AND IDEATION

Setting Up and Using PocketSmart AI for Budget Planning

What You'll Learn

- How to configure PocketSmart AI with a Google Gemini API key and run it locally in Visual Studio Code
- How to register an account and sign in to the PocketSmart AI web app
- How to use the Home, Party, and Jewelry planners to generate AI-powered budget recommendations
- How to browse suggested products on external retail sites like IKEA, Amazon, and Flipkart
- How to navigate the dashboard, review your plan history, and log out securely

Introduction

This guide walks you through setting up PocketSmart AI — a budget-planning tool powered by Google's Gemini AI — and using its three planners (Home, Party, and Jewelry) to generate budget-aware recommendations. It's intended for anyone installing PocketSmart AI for the first time or learning how to use its planning features day to day.

Prerequisites

Before you begin, make sure you have:

- Visual Studio Code installed, with the [PocketSmartAI_FIXED_VSCode](#) project folder available
- A Python virtual environment already set up in the project ([venv](#) folder)
- A Google account to access Google AI Studio and generate an API key
- A web browser for accessing Google AI Studio and the PocketSmart AI web app

02 REQUIREMENT ANALYSIS

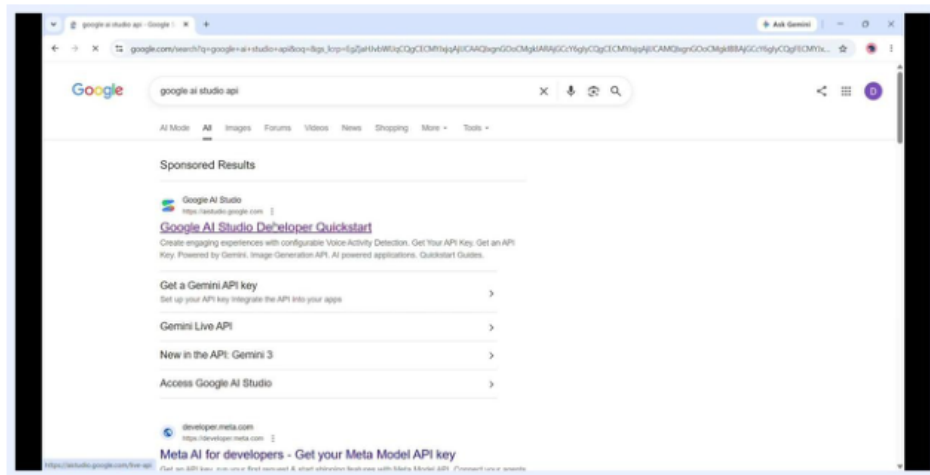
Procedure: Setting up PocketSmart AI with a Gemini API key

Step 1: Open the project in Visual Studio Code

Launch Visual Studio Code and open the [PocketSmartAI_FIXED_VSCode](#) project folder. In the Explorer sidebar, open the [.env.example](#) file to review the environment variables you'll need to configure:

- [APP_NAME=PocketSmart AI](#)
- [SECRET_KEY=change-this-to-a-long-random-secret](#)
- [DATABASE_URI=sqlite:///pocketsmart.db](#)
- [GEMINI_API_KEY=](#)
- [GEMINI_MODEL=gemini-2.5-flash](#)
- [ACCESS_TOKEN_EXPIRE_MINUTES=120](#)
- [MAX_IMAGE_SIZE_MB=5](#)
- [ALLOWED_ORIGINS=http://127.0.0.1:8000,http://localhost:8000](#)

The workspace includes folders such as [venv](#) and [app](#), along with files like [README.md](#) and [pocketsmart.db](#).

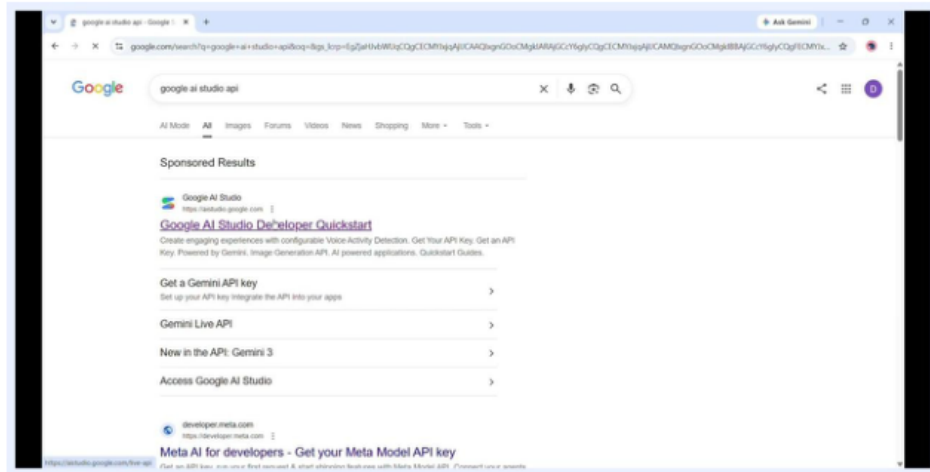


[.env.example](#) file with environment variables

02 REQUIREMENT ANALYSIS

Step 2: Search for Google AI Studio

Open a web browser, go to Google, and search for "google ai studio api." The top result is typically the Google AI Studio Developer Quickstart at aistudio.google.com, alongside related links such as "Get a Gemini API key" and "Access Google AI Studio."



Google search results for Google AI Studio API

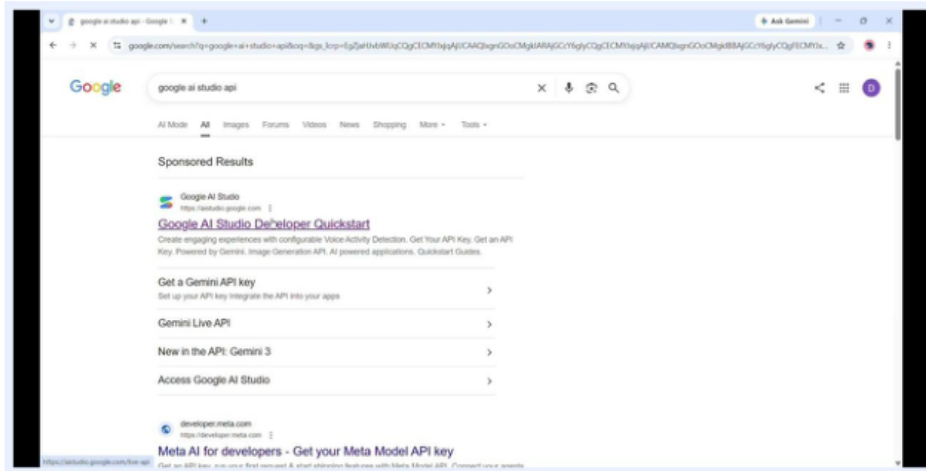
Step 3: Access API keys in Google AI Studio

Click through to Google AI Studio and sign in if prompted. Go to the API Keys section in the left sidebar. Here you'll see:

- A list of existing API keys grouped by key or project
- Each key's name, associated project (e.g., [Default Gemini Project](#)), creation date, and billing tier

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- A Create API key button in the upper right



Google AI Studio API Keys page

Step 4: Copy the Gemini API key

Click Create API key to generate a new one, or select an existing Gemini API key from the list. Copy the key value — you'll need it in the next step.

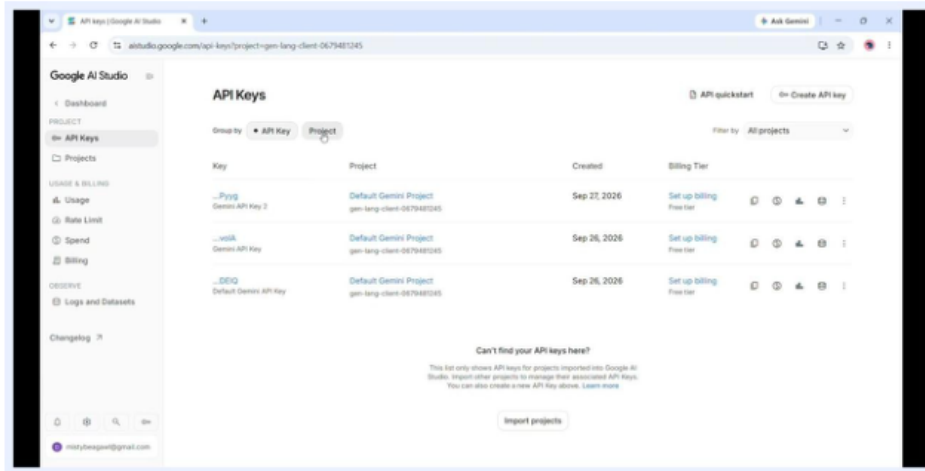
Step 5: Add the API key to your .env file

Return to Visual Studio Code and paste the copied key into the `GEMINI_API_KEY` field in your `.env` (or `.env.example`) file:

```
GEMINI_API_KEY=AQ.AB8Rn6JqfUqR2yLfn6fM4ZUzSWtNoFpkOn1GZSHsNHbYrTw
```

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Confirm the other environment variables are set as needed before continuing.



.env file with Gemini API key filled in

Step 6: Activate the Python virtual environment

Open the integrated terminal in VS Code and activate the virtual environment:

- On Windows: `.\venv\Scripts\Activate.ps1`

Your terminal prompt should now show the environment is active, indicated by `(venv)`.

Step 7: Start the application

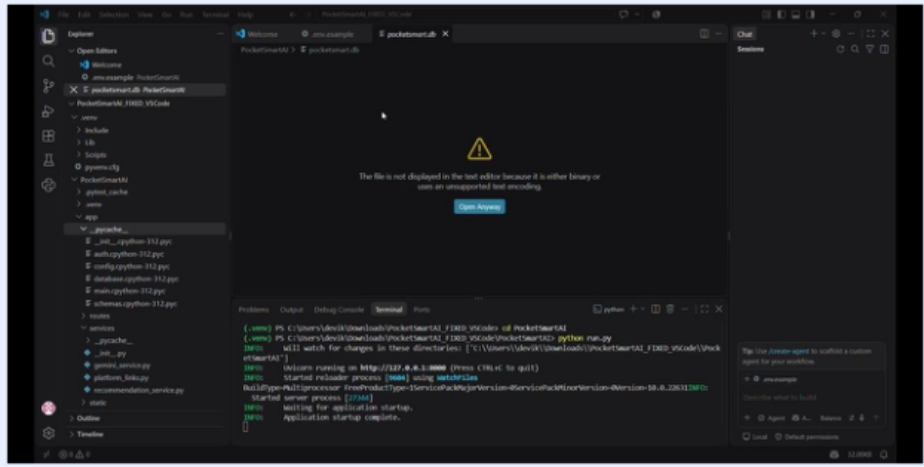
Navigate to the project directory if needed (`cd PocketSmartAI`), then start the app:

```
python run.py
```

You should see confirmation in the terminal output:

```
INFO: Uvicorn running on http://127.0.0.1:8000
INFO: Application startup complete
```

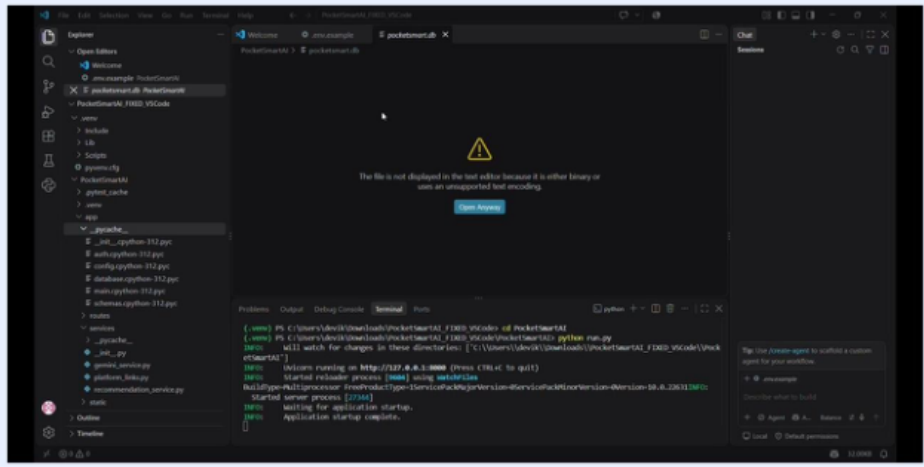
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Terminal showing virtual environment activation and app startup

Step 8: Verify the database and application status

In the Explorer sidebar, you'll see the `pocketSMART.db` file. If you try to open it, VS Code shows a warning that the file can't be displayed as text — this is expected, since it's a binary SQLite database file. The terminal should continue showing logs confirming the app is running.



Binary file warning for pocketsmart.db and terminal logs

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Note: The "file is not displayed" warning for [pocketsmart.db](#) is normal — SQLite database files are binary and aren't meant to be opened as text.

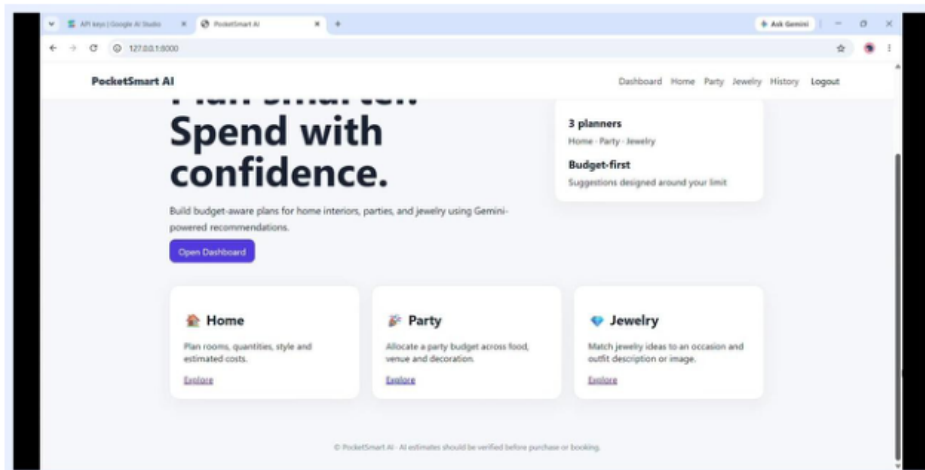
You're ready to move on to using the web app itself.

Procedure: Registering an account

Step 1: Access the PocketSmart AI web app

Open your browser and go to <http://127.0.0.1:8000>. The homepage includes:

- A navigation bar with links to Dashboard, Home, Party, Jewelry, History, and Logout
- The headline "Plan smart. Spend with confidence."
- A description of the app's purpose: building budget-aware plans for home interiors, parties, and jewelry using Gemini-powered recommendations
- An Open Dashboard button
- A summary box highlighting "3 planners" and "Budget-first" suggestions
- Three planner cards — Home, Party, and Jewelry — each with an Explore link



PocketSmart AI homepage with planner options



Note: A footer reminder on the homepage states that AI estimates should be verified before purchase or booking.

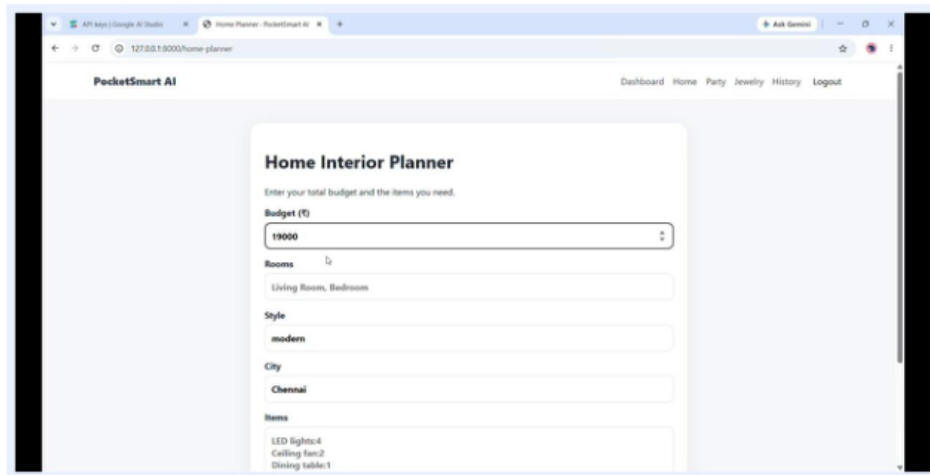
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Step 2: Register a new account

Click Register in the top right corner. Fill in the Create account form:

- Username: [sai](#)
- Email: [devipriya23@gmail.com](#)
- Password: choose a secure password

Click Register to create your account.

A screenshot of a web browser displaying the 'Home Interior Planner' form. The browser's address bar shows '127.0.0.1:8000/home-planner'. The page has a navigation bar with links: 'Dashboard', 'Home', 'Party', 'Jewelry', 'History', and 'Logout'. The form itself is titled 'Home Interior Planner' and includes a sub-header 'Enter your total budget and the items you need.' It contains several input fields: 'Budget (₹)' with the value '19000', 'Rooms' with the value 'Living Room, Bedroom', 'Style' with the value 'modern', and 'City' with the value 'Chennai'. There is also a section for 'Items' which lists 'LED lights:4', 'Ceiling fans:2', and 'Dining table:1'.

Registration form with username, email, and password fields

That's all you need to get started — you're ready to try the planners.

Procedure: Using the Home Interior planner

Step 1: Fill in the Home Interior Planner form

Navigate to the Home planner from the homepage or navigation bar. Fill in the form:

- Budget (₹): [12000](#)
- Rooms: [Living Room, Bedroom](#)
- Style: [modern](#)
- City: [Chennai](#)

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- Items: led lights 4, ceiling fan 3

The screenshot shows a web browser window with the URL '127.0.0.1:8000/home-planner'. The page title is 'PocketSmart AI'. The navigation bar includes links for 'Dashboard', 'Home', 'Party', 'Jewelry', 'History', and 'Logout'. The main form has the following fields: 'Budget (₹)' with the value '12000', 'Rooms' with the value '3', 'Style' with the value 'modern', 'City' with the value 'Chennai', and 'Items' with the value 'led lights 4, ceiling fan 3'. A blue button labeled 'Generate Recommendations' is located at the bottom of the form.

Home Interior Planner form filled with budget, rooms, style, and items

Step 2: Generate recommendations

Click Generate Recommendations at the bottom of the form. The app processes your inputs and returns AI-powered suggestions based on your budget and requirements.

Step 3: Review your recommendations

The results page displays a budget summary and recommended items.



Note: If a live AI response isn't available, PocketSmart AI shows fallback recommendations instead, with a notice explaining this.

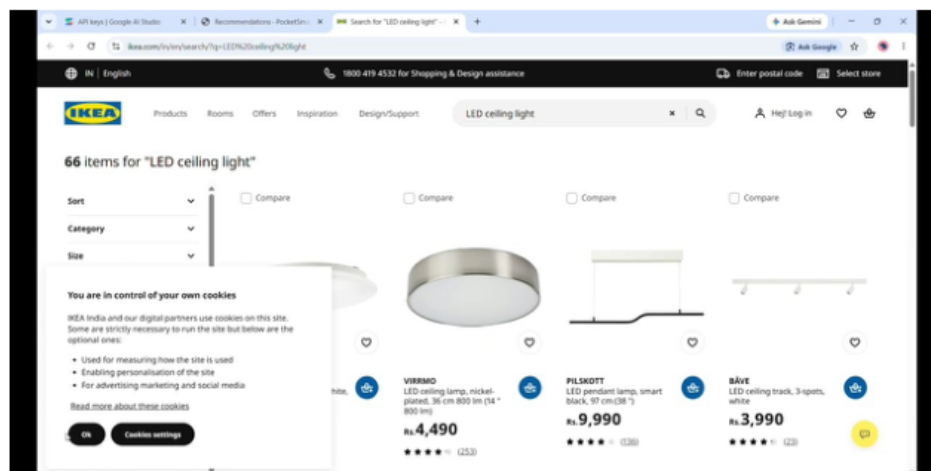
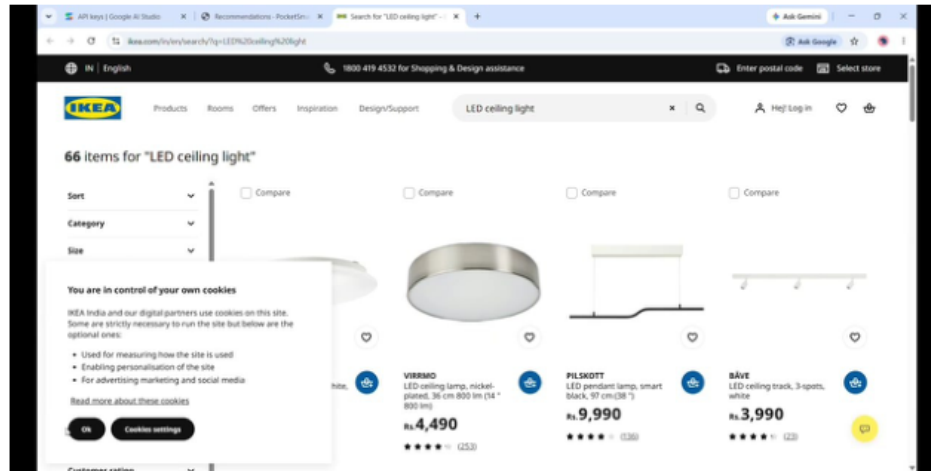
Example results:

- Estimated total: ₹4,000
- Remaining: ₹8,000
- LED ceiling light — Lighting, IKEA, ₹1,200 — "Energy-efficient and suitable for general room lighting"
- Ceiling fan — Fan, Amazon, ₹2,800 — "A practical choice for airflow while keeping the budget controlled"

Each item includes a Search on platform link that opens the retailer's site with a matching search. You can also click Open saved result to view or manage saved

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recommendations, which shows the same fallback notice, budget summary, and itemized breakdown with direct search links.



Tip: Prices shown are estimates — use the platform links to check current availability and pricing before you buy.

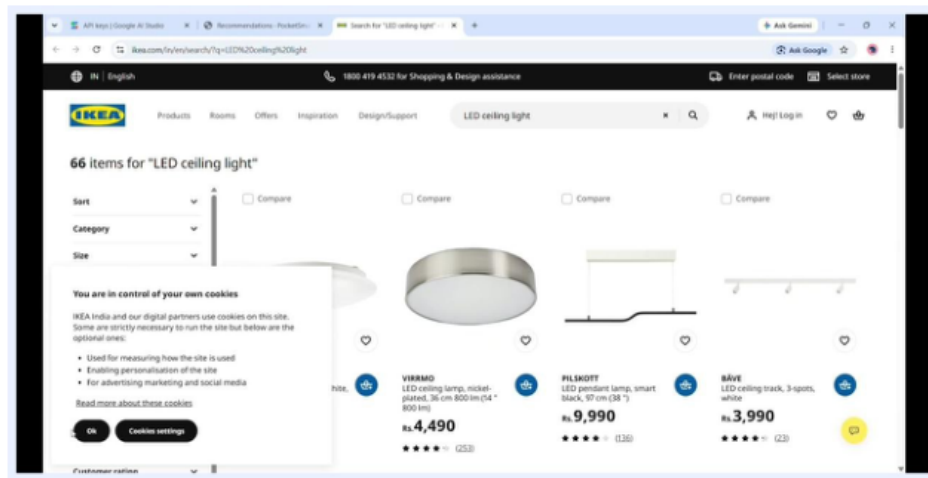
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Step 4: Browse products on external platforms

Click Search on platform for an item (for example, the LED ceiling light) to open a new tab on the retailer's site. On IKEA, you'll see multiple product options with images, names, prices, and ratings, such as:

- VIRRMO LED ceiling lamp, nickel-plated, 36 cm, 800 lm — Rs. 4,490
- PILSKOTT LED pendant lamp, smart black, 97 cm — Rs. 9,990
- BÄVE LED ceiling track, 3-spots, white — Rs. 3,990

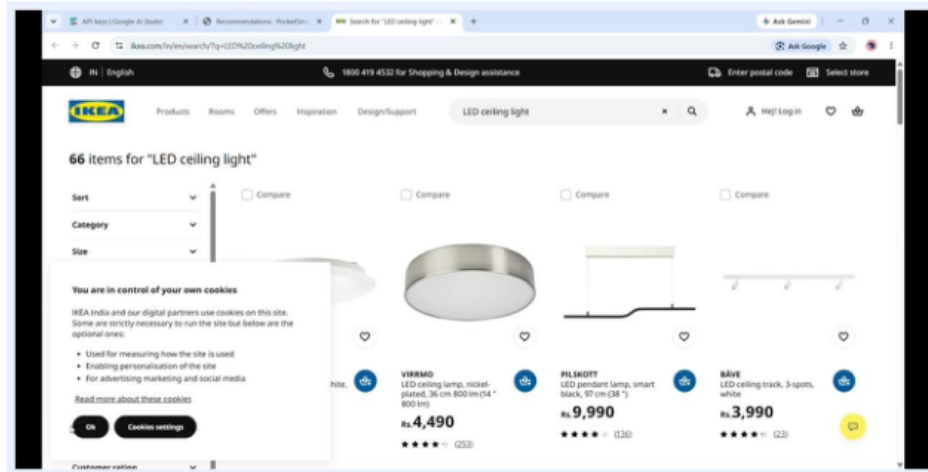
A cookie consent dialog may appear — click Ok or adjust settings as needed.



IKEA search results for LED ceiling light

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Return to the PocketSmart AI Recommendation Details tab to continue reviewing other items, such as clicking Search on platform for the ceiling fan to open Amazon.



Recommendation details page with search links for each item

Procedure: Planning a party budget

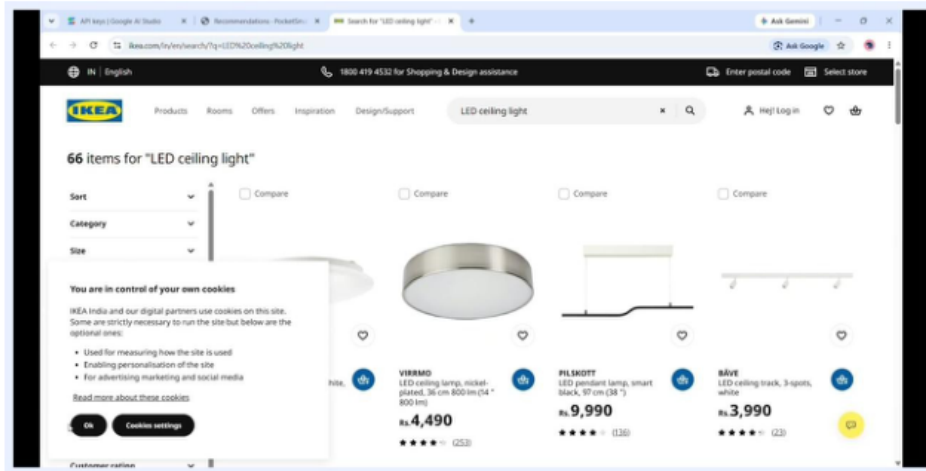
Step 1: Fill in the Party Budget Planner form

Navigate to the Party section from the navigation bar or planner cards. The form includes:

- Total Budget (₹)
- Guest Count
- Event Type (e.g., Birthday / Corporate / Wedding)
- Venue Details
- Food Preference

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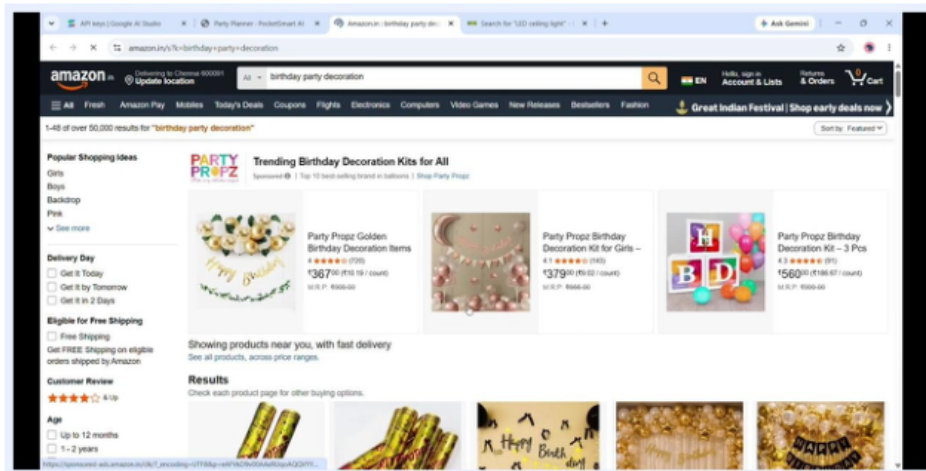
- City



Party Budget Planner form fields

Enter your details, for example:

- Total Budget (₹): 10000
- Guest Count: 20
- Event Type: birthday
- Venue Details: chennai outdoor
- Food Preference: mixed
- City: Chennai



Party Budget Planner form fully completed